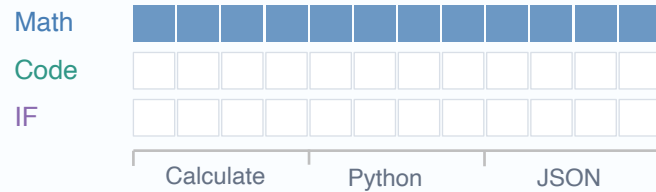


A Rethinking routing

“Sum 1...100; verify in Python;
return the answer in JSON.”

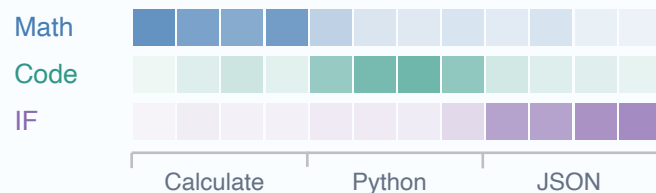
Prompt-level hard routing

Domain label → one teacher throughout

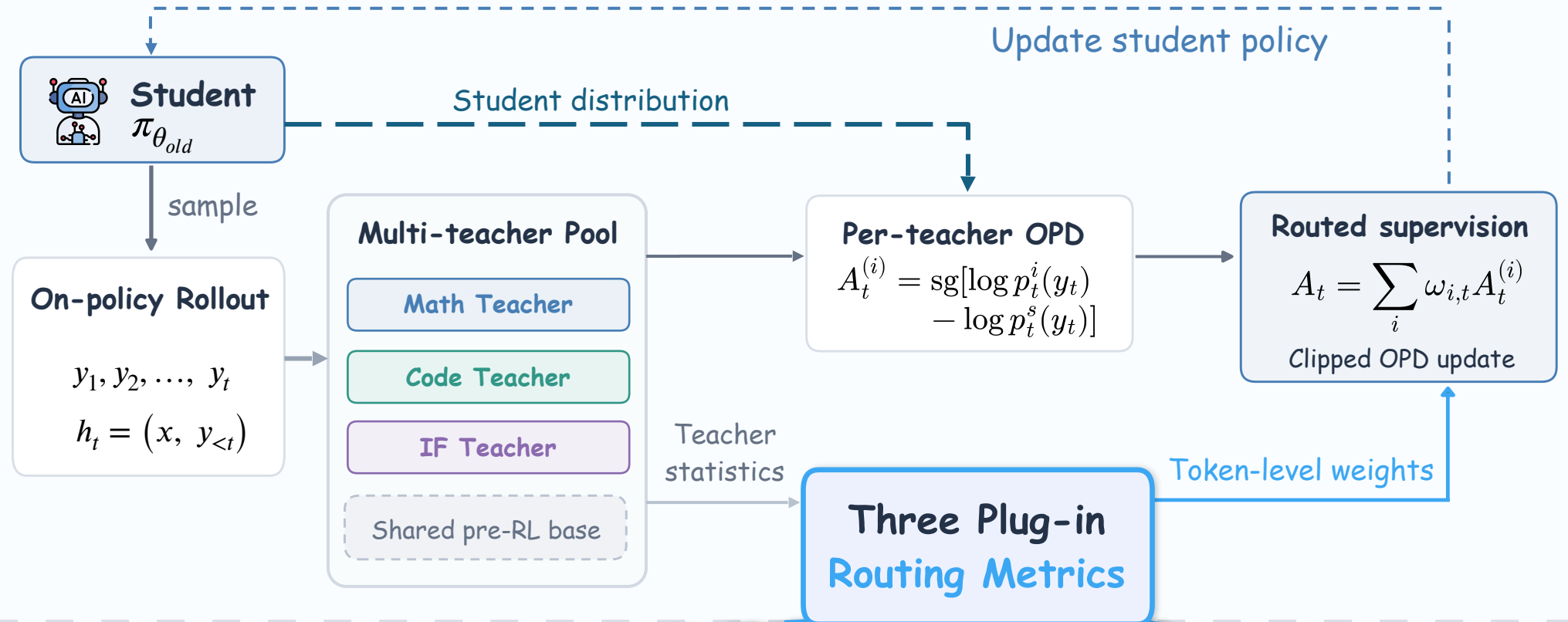


Token-level metric routing

All teachers; weights vary with context



B Multi-teacher on-policy distillation



Entropy (Teacher confidence)

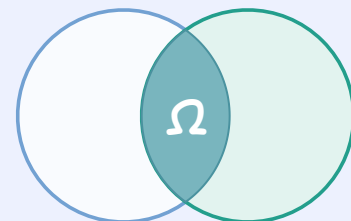


$$-\mathcal{H}(p_i^t)$$

Lower entropy → higher weights
SoftMax over teacher

Novelty (Accessible teacher-student difference)

Student top-k Teacher top-k



Shared candidates

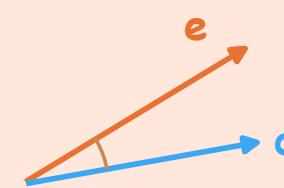
$$\Omega_{i,t} = \text{TopK}(p_t^S) \cap \text{TopK}(p_t^i)$$

$$a_{i,t} = \sqrt{p_t^S(\Omega) \cdot p_t^i(\Omega)}$$

$$s_{i,t} = a_{i,t} [1 - e^{-\text{JS}_{\Omega}/\tau_N}]$$

Higher distribution difference
→ higher weights; SoftMax over teacher

ExpertAlign (Specialization-aligned supervision)



Compare directions on
student top-k support

$$\mathbf{e}_{i,t} = \log p_t^i - \log p_t^{\text{base}}$$

$$\mathbf{d}_{i,t} = \log p_t^i - \log p_t^S$$

$$c_{i,t} = \max(\cos(\mathbf{e}_{i,t}, \mathbf{d}_{i,t}), 0)$$

Higher relative alignment → higher weights
Gate positive alignment; weight by cosine